Alert Triage Practice

---

1. Introduction to Alert Triage

What is Alert Triage?

Alert triage is the process of reviewing, validating, prioritizing, and investigating security alerts generated by security tools such as SIEMs, IDS/IPS systems, EDR solutions, and firewalls.

The primary objective of alert triage is to determine:

- Whether the alert is genuine.
- The severity of the threat.
- The potential impact.
- Required response actions.

Effective triage helps SOC analysts focus on real threats while reducing time spent on false positives.

---

2. Alert Triage Workflow

Step 1: Receive Alert

Security tools generate alerts based on suspicious activity.

Example:

Alert: Multiple SSH Login Failures
Source IP: 192.168.1.100
Target: Linux Server

Step 2: Verify Alert

Analysts validate:

- Source IP
- Destination IP
- Username
- Timestamp
- Associated logs

Step 3: Determine Severity

Severity is assigned based on:

- Asset importance

- Threat type
- Business impact
- Known vulnerabilities

Step 4: Investigate

Analyst gathers evidence:

- System logs
- Authentication records
- Network traffic
- Threat intelligence

Step 5: Classify

Classification options:

- True Positive
- False Positive
- Benign Activity
- Requires Further Investigation

Step 6: Escalate or Close

Depending on findings:

- Escalate to Incident Response Team
- Continue monitoring
- Close alert

---

3. Triage Simulation – Brute Force SSH Attack

Alert Information

Field	Value
Alert ID	ALT-002
Alert Type	SSH Brute Force
Source IP	192.168.1.100

Field	Value
Destination IP	192.168.1.10
Severity	Medium
Status	Open

Alert Description

The SIEM detected 150 failed SSH login attempts against a Linux server within 5 minutes.

Investigation Steps

Step 1

Review authentication logs.

Finding:

Failed password for root
Failed password for admin
Failed password for test

Step 2

Count failed attempts.

Result:

150 failures within 5 minutes.

Step 3

Check source reputation.

IP Investigated:

192.168.1.100

Step 4

Determine impact.

No successful login detected.

Final Assessment

Classification:

True Positive

Priority:

Medium

Recommended Action:

- Block source IP
- Enable account lockout
- Increase monitoring

---

4. Threat Intelligence Validation

IOC Validation Process

Threat Intelligence platforms help determine whether an indicator is malicious.

Common Platforms:

- VirusTotal
- AlienVault OTX
- AbuseIPDB
- Cisco Talos

Sample Validation

IOC

IP Address:

185.220.101.25

AlienVault OTX Results

Findings:

- Known malicious IP
- Associated with botnet activity
- Reported multiple times

Risk Assessment

Risk Level:

High

Analyst Conclusion

The source IP has a malicious reputation and should be blocked immediately.

---

5. False Positive Validation

Alert Information

Alert:

Port Scan Detected

Investigation

Source:

Internal Vulnerability Scanner

IP:

192.168.1.50

Findings

The scan was initiated by authorized security personnel.

Classification

False Positive

Action Taken

No action required.

Alert closed.

---

6. Triage Documentation Sheet

Alert ID	Description	Source IP	Priority	Status
ALT-001	Phishing Email	10.0.0.5	High	Closed
ALT-002	SSH Brute Force	192.168.1.100	Medium	Open

Alert ID	Description	Source IP	Priority	Status
ALT-003	Malware Detection	172.16.0.20	High	Investigating
ALT-004	Port Scan	192.168.1.50	Low	Closed
ALT-005	Data Exfiltration	203.0.113.25	Critical	Escalated

---

7. Triage Decision Matrix

Condition	Classification
Malicious activity confirmed	True Positive
Legitimate activity triggered alert	False Positive
Insufficient evidence	Further Investigation
Normal business activity	Benign Event

---

8. SOC Analyst Investigation Notes

Alert ID

ALT-005

Alert Type

Possible Data Exfiltration

Observations

- Large outbound traffic detected.
- Data transferred outside business hours.
- Unknown destination IP identified.

Investigation Performed

- Reviewed firewall logs.
- Verified user activity.
- Checked destination reputation.
- Examined endpoint logs.

Findings

Unauthorized transfer of sensitive files suspected.

Recommendation

Immediate escalation to Incident Response Team.

Priority set to Critical.

---

9. Summary Report

During alert triage exercises, multiple alerts were analyzed including phishing attempts, SSH brute-force attacks, malware detections, and data exfiltration activities. Threat intelligence platforms were used to validate indicators of compromise and determine alert legitimacy. Alerts were classified as true positives, false positives, or requiring further investigation. Appropriate response actions such as blocking malicious IPs, escalating incidents, and closing benign alerts were documented. The exercise improved understanding of SOC alert handling and prioritization procedures.